Ritesh Mamidi

Applied AI Engineering

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SUMMARY

Data and AI quality professional focused on grounded retrieval, evaluation and controlled tool workflows. Builds independent Python systems with typed APIs, versioned knowledge and tested control boundaries. Separates deterministic baseline behavior from language-model performance and uses reproducible failure analysis to guide implementation decisions.

TECHNICAL SKILLS

Python, SQL, pandas; FastAPI, Pydantic, REST APIs, SQLite, Git; AI/ML output evaluation, RAG/vector retrieval, deterministic agent workflows, structured outputs, human review; ETL, data validation, KPI analysis, Power BI, Tableau; requirements, root-cause analysis and stakeholder communication.

PROFESSIONAL EXPERIENCE

Apple Maps Client Engagement — Product & AI Quality Operations | Sep 2025–Present

- Validate geospatial data and AI/ML outputs against structured policy and quality rubrics; investigate incomplete sources, edge cases and recurring failure patterns.

- Support QA/QC checks, KPI monitoring and operational reporting with Python, SQL and Power BI; document findings for cross-functional partners.

- Analyze workflows, conduct quality-control reviews and support training to improve consistency. Client engagement; does not imply direct Apple employment.

Heart Hospitality LLC — Business / Operations Analyst | Jan 2024–Aug 2025

- Built Python/SQL ETL workflows and automated recurring reporting using Power BI and Excel KPI dashboards.

- Analyzed operational and customer data, built forecasting/classification models and designed relational data models to support planning.

- Gathered stakeholder requirements, documented processes and aligned analytics with business goals.

Liberated Monks Pvt. Ltd. — Data / Business Analyst | Jan 2022–Dec 2023

- Built Python/SQL ETL workflows and Tableau dashboards for acquisition, engagement and retention analysis.

- Maintained data validation rules and KPI definitions; translated cross-team requirements into analyses and evaluated segmentation models.

INDEPENDENT ENGINEERING — SYNTHETIC DEMONSTRATIONS

- Built five local simulated workflows spanning document RAG, controlled tools, authenticated loopback HTTP, CSV reconciliation and configuration/checkpoint rollback.

- Implemented operator/reviewer separation, tenant-scoped data, stable idempotency keys, durable receipts and audit trails; tested bad input, denied actions, uncertain outcomes and recovery.

- Validated 57 API/control tests, eight deterministic retrieval fixtures and five end-to-end local demos; supplied architecture, reproduction instructions and recorded execution. No production deployment or live-model quality is claimed.

EDUCATION

M.S., Business Analytics — St. Francis College | May 2025

M.Tech (Integrated), Software Engineering — Vellore Institute of Technology (VIT) | May 2023